

Pareto ladder (tightened)

How hard can we compress the LatentMAS relay?

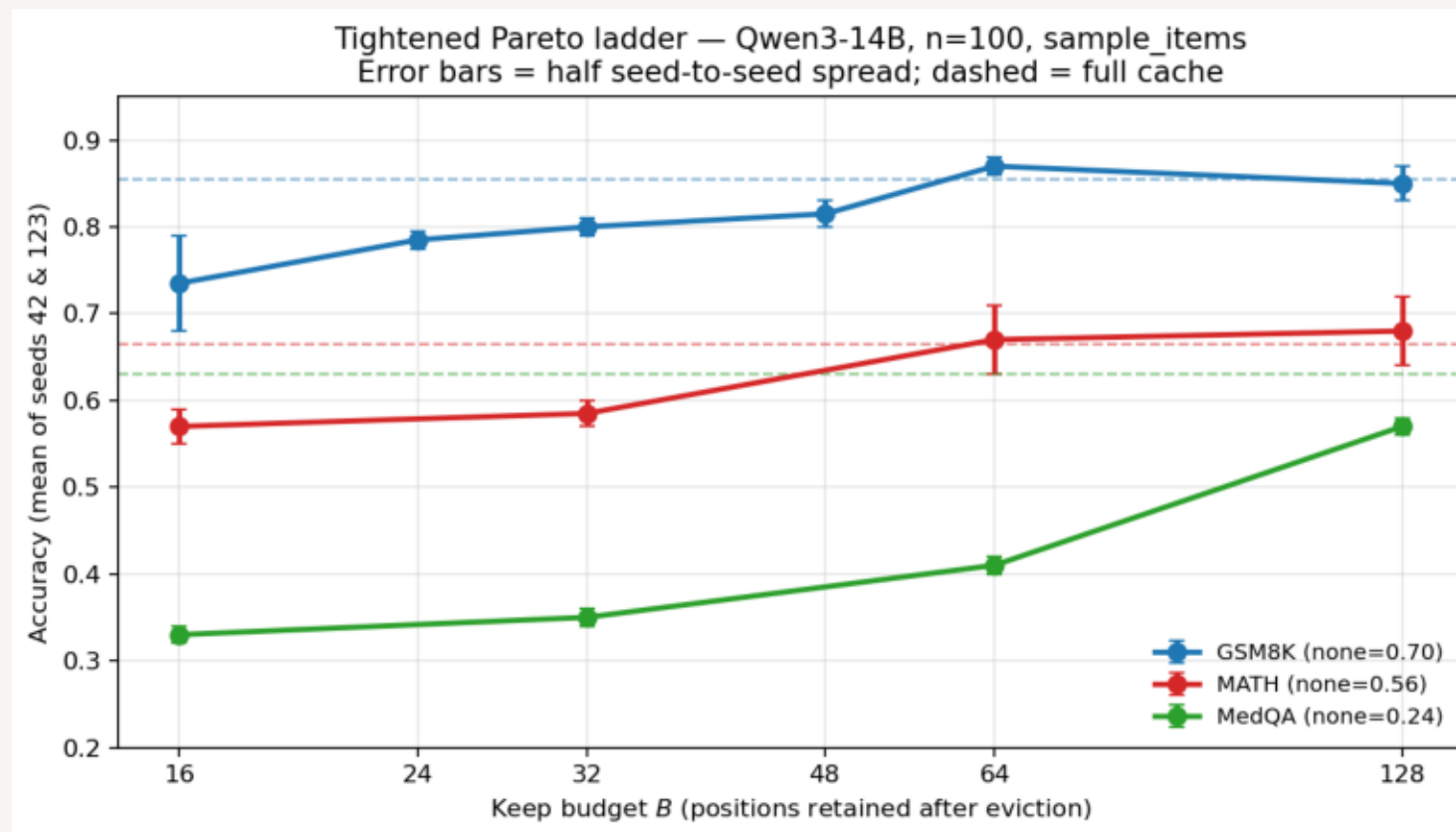
Qwen3-14B · n=100 · seeds 42 & 123 (sample_items) · bootstrap CIs

One upstream build → plain eviction at each B → Judger decode
Greedy · no ASC (memory × accuracy)

- LatentMAS relay KV is largely content-insensitive → redundant scaffold
- Systems consequence: shrink it before the Judger (sink + key-norm top-k)
- Question: for each task, how small can B get before accuracy drops?
- This deck: $n=100$, two real item-subsets, CIs

Paper-shaped claim

A task-conditional memory-accuracy Pareto — not “40× free everywhere.”



- GSM8K: soft at $B=16$; clean iso by $B \approx 64$ ($\sim 10\times$); $B=32$ still mild tax
- MATH: free-ish from $B \approx 64$ -128 (~ 5 -10 $\times$)
- MedQA: still -6 pp at $B=128$ — thickness required; none collapses hard

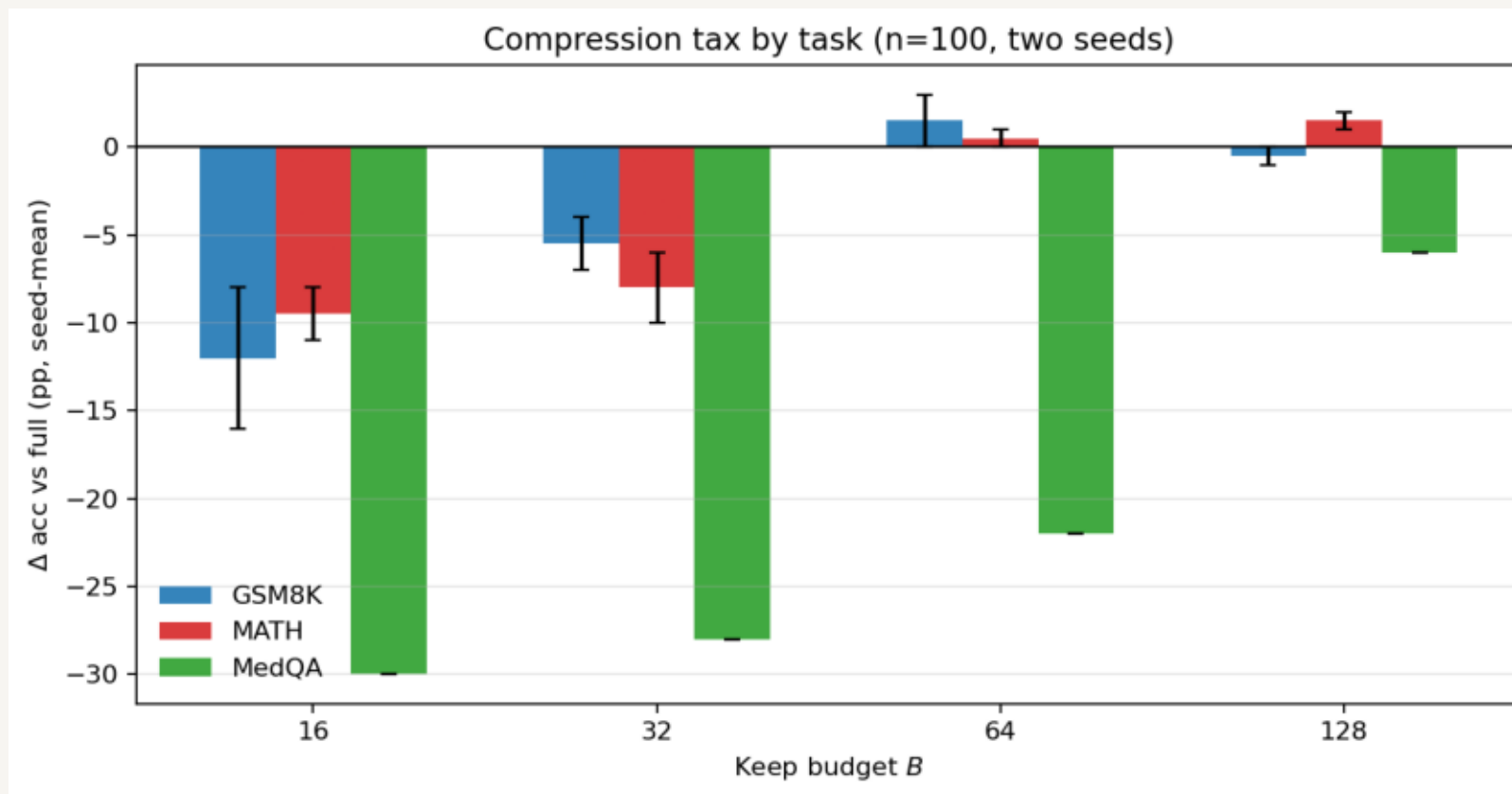
Task	full	B16	B32	B64	B128	none
GSM8K	0.855	0.735	0.800	0.870	0.850	0.700
MATH	0.665	0.570	0.585	0.670	0.680	0.555
MedQA	0.630	0.330	0.350	0.410	0.570	0.240

Recommended B per task

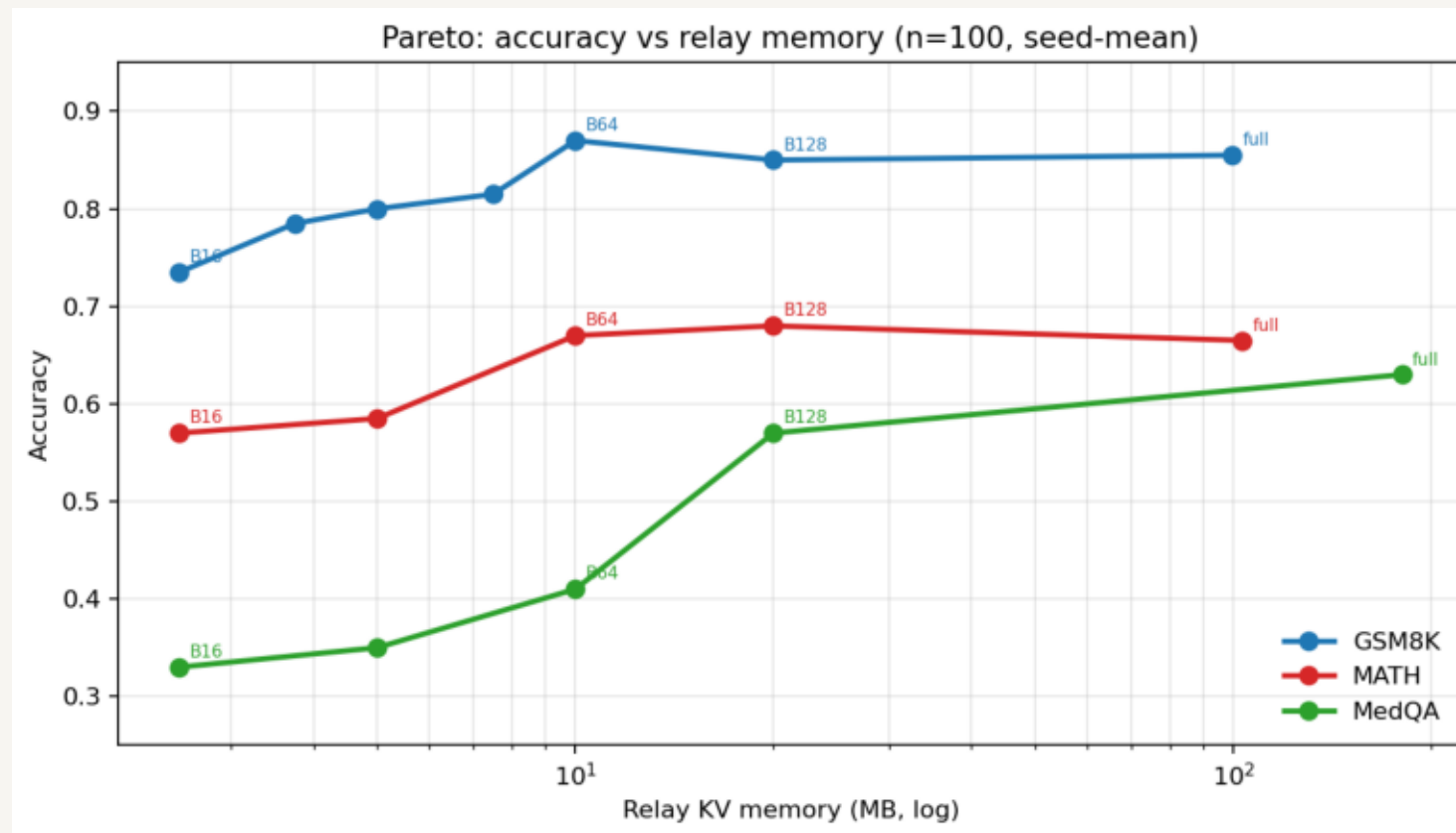
- GSM8K: B=64 (~10×) for iso-acc
- MATH: B=64-128 (~5-10×)
- MedQA: ≥ 128 ; 16-64 collapse

Presence still matters

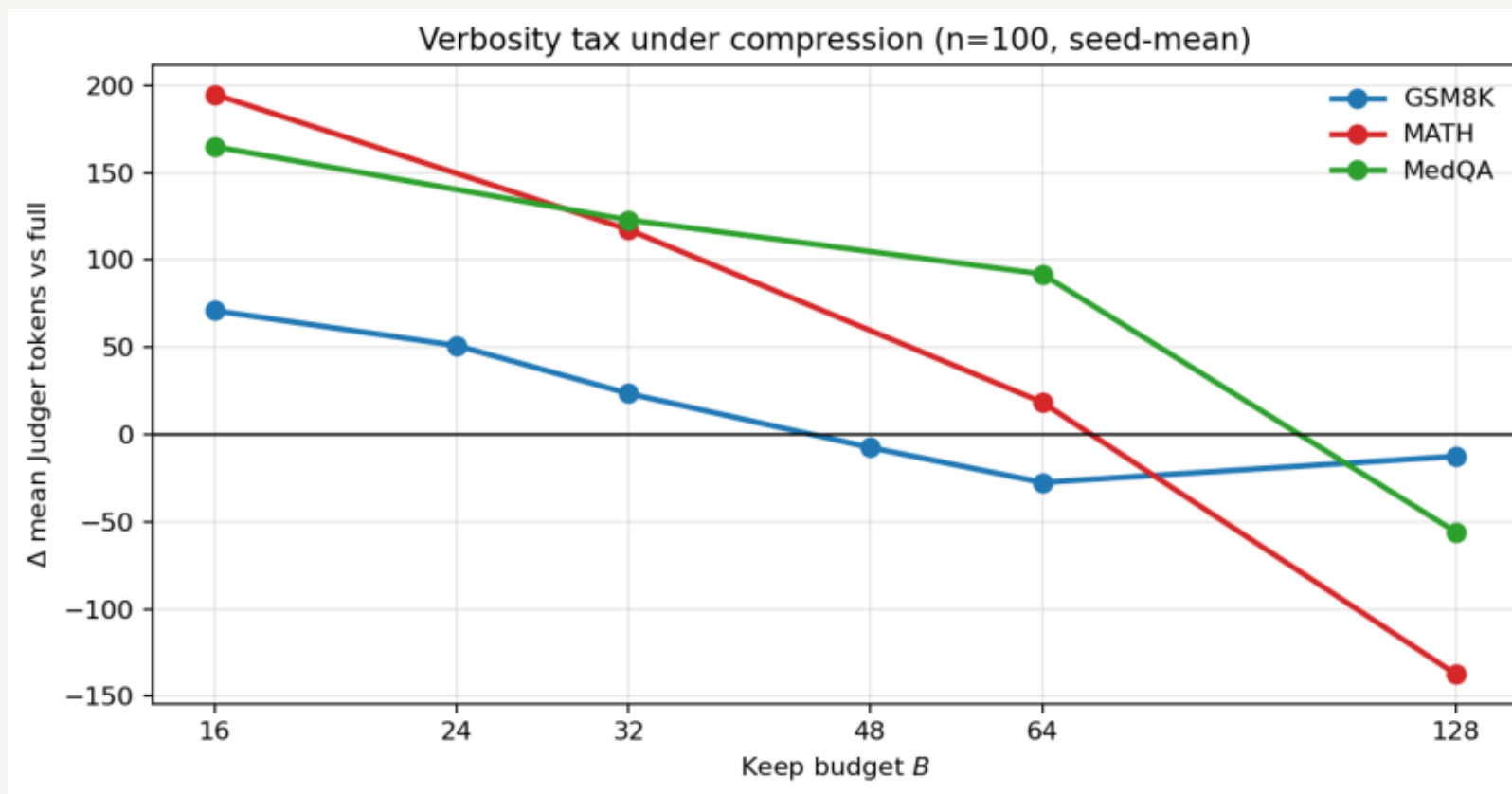
none $\ll$ full on every task.
Compression trims redundancy;
it does not delete the scaffold.



- Easy task $\Rightarrow$ flat tax until extreme B
- Harder tasks $\Rightarrow$ tax grows as B shrinks
- Same compressor, different residual — task-conditional keep



- Full relays: ~100 MB (GSM8K/MATH) vs ~180 MB (MedQA)
- B=16 → 2.5 MB (~40-70×); only GSM8K approaches the frontier later
- Sell recommended B per task / Pareto, not one universal ratio



- Aggressive shrink lengthens Judger CoT (esp. MATH)
- Token tax fades as B grows

- Mechanism: relay is a scaffold, not an instance-specific channel
- Systems: compress hard on easy tasks; keep thicker residual on hard ones
- Tightened evidence: $n=100$, two real subsets, Cls — story unchanged, sharper knees
- GSM8K iso @ $B \approx 64$ ($\sim 10\times$); MATH @ 64–128; MedQA still taxed at 128